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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 60.14

$$\begin{aligned}
 & \int \frac{6x+2-\sqrt{4x+1}}{2x+1+\sqrt{4x+1}} dx \\
 & u = \sqrt{4x+1} \Rightarrow x = \frac{u^2-1}{4} \Rightarrow dx = \frac{u}{2} du \\
 & = \int \frac{\frac{3u^2+1}{2} - u}{\frac{u^2+1}{2} + u} \cdot \frac{u}{2} du \\
 & = \int \frac{\frac{3u^2-2u+1}{2}}{\frac{u^2+2u+1}{2}} \cdot \frac{u}{2} du \\
 & = \int \frac{3u^2-2u+1}{u^2+2u+1} \cdot \frac{u}{2} du = \frac{1}{2} \int \frac{3u^3-2u^2+u}{(u+1)^2} du \\
 & u = t+1 \Rightarrow du = dt \\
 & = \frac{1}{2} \int \frac{3(u-1)^3-2(u-1)^2+(u-1)}{u^2} du \\
 & = \frac{1}{2} \int \frac{3(u^3-3u^2+3u-1)-2(u^2-2u+1)+u-1}{u^2} du \\
 & = \frac{1}{2} \int \frac{3u^3-11u^2+14u-6}{u^2} du \\
 & = \frac{1}{2} \int \left(3u-11+\frac{14}{u}-\frac{6}{u^2} \right) du \\
 & = \frac{1}{2} \left(\frac{3u^2}{2} - 11u + 14 \ln|u| + \frac{6}{u} \right) + C \\
 & = \frac{3(t+1)^2}{4} - \frac{11(t+1)}{2} + 7 \ln|u+1| + \frac{3}{u+1} + C \\
 & = \frac{3(\sqrt{4x+1}+1)^2}{4} - \frac{11(\sqrt{4x+1}+1)}{2} + 7 \ln|\sqrt{4x+1}+1| + \frac{3}{\sqrt{4x+1}+1} + C
 \end{aligned}$$

References